LAPORAN PRAKTIKUM
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)
SEMESTER III



Acara 16
RESTful API + endpoint requestresponse

NAMA :

ANGELLO ERIYA DEWANGGA (E41250776)

PROGRAM STUDI DILUAR KAMPUS UTAMA (PSDKU)

TEKNIK INFORMATIKA KAB. NGANJUK

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

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A. Tujuan Pengujian

Pengujian dilakukan untuk memastikan endpoint API mahasiswa dapat menjalankan operasi GET, POST, PUT, dan DELETE dengan benar serta menerapkan autentikasi JSON Web Token (JWT) pada endpoint yang memerlukan akses terproteksi.

Pengujian dilakukan menggunakan **curl.exe** melalui terminal PowerShell sebagai alternatif dari Postman.

1. Metode GET

```
$base = "http://localhost/BKPM/Acara16/si-
akademik/api/mahasiswa"
curl.exe -i $base
```

2. Metode POST

```
$body = '{"nim":"2024004","nama":"Budi
Uji","email":"budiuji@mail.com","jurusan":"Teknik
Informatika"}'
[System.IO.File]::WriteAllText(
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X POST $base `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

3. Metode PUT

```
$body = '{"nim":"2024001","nama":"Budi Santoso
Updated","email":"budi@mail.com","jurusan":"Sistem
Informasi"}'
[System.IO.File]::WriteAllText(
```

```
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X PUT "${base}?id=1" `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

4. Metode DELETE

```
curl.exe -i -X DELETE "${base}?id=1"
```

- **$base** digunakan untuk menyimpan URL endpoint API mahasiswa agar dapat digunakan kembali pada perintah berikutnya.
- **http://localhost** menunjukkan bahwa aplikasi dijalankan pada komputer lokal.
- **/BKPM/Acara16/si-akademik/api/mahasiswa** merupakan alamat endpoint yang menangani data mahasiswa

- **curl.exe** digunakan untuk mengirim permintaan HTTP ke server.
- **-X POST** menentukan bahwa permintaan HTTP menggunakan metode POST untuk mengirim data baru ke API.
- **-X PUT** menentukan bahwa permintaan HTTP digunakan untuk memperbarui data.
- **-X DELETE** menentukan bahwa permintaan HTTP digunakan untuk menghapus data.

- **`-i`** digunakan untuk menampilkan header respons HTTP beserta isi respons dari server.
- **-H "Content-Type: application/json**" memberi tahu server bahwa isi permintaan menggunakan format JSON.
- **--data-binary "@body.json"** mengirim isi file body.json sebagai badan permintaan (request body).

B. Hasil Pengujian

```
PS C:\laragon\www\BKPM\Acara16> $base = "http://localhost/BKPM/Acara16/si-akademik/api/mahasiswa"
PS C:\laragon\www\BKPM\Acara16> curl.exe -i $base
HTTP/1.1 200 OK
Date: Tue, 20 Oct 2026 14:40:04 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Content-Length: 777
Content-Type: application/json; charset=utf-8

{
    "status": "success",
    "total": 3,
    "data": [
        {
            "id": 1,
            "nim": "2024001",
            "nama": "Budi Santoso",
            "email": "budi@mail.com",
            "jurusan": "Teknik Informatika",
            "created_at": "2026-10-20 20:23:25"
        },
        {
            "id": 2,
            "nim": "2024002",
            "nama": "Siti Aminah",
            "email": "siti@mail.com",
            "jurusan": "Sistem Informasi",
            "created_at": "2026-10-20 20:23:25"
        },
        {
            "id": 3,
            "nim": "2024003",
            "nama": "Andi Pratama",
            "email": "andi@mail.com",
            "jurusan": "Teknik Informatika",
            "created_at": "2026-10-20 20:23:25"
        }
    ]
}
```

1. Menampilkan seluruh data mahasiswa menggunakan GET

```
PS C:\laragon\www\BKPM\Acara16> curl.exe -i "$base/1"
HTTP/1.1 200 OK
Date: Tue, 20 Oct 2026 14:41:06 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Content-Length: 242
Content-Type: application/json; charset=utf-8

{
    "status": "success",
    "data": {
        "id": 1,
        "nim": "2024001",
        "nama": "Budi Santoso",
        "email": "budi@mail.com",
        "jurusan": "Teknik Informatika",
        "created_at": "2026-10-20 20:23:25"
    }
}
```

2. Menampilkan data mahasiswa berdasarkan ID

```
PS C:\laragon\www\BKPM\Acara16> '{"nim":"2024004","nama":"Dewi Lestari","email":"dewi@mail.com","jurusan":"Sistem Informasi"}' | Set-Content body.json -Encoding ascii -NoNewline
>> curl.exe -i -X POST $base -H "Content-Type: application/json" --data-binary "@body.json"
HTTP/1.1 201 Created
Date: Tue, 20 Oct 2026 14:44:30 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Location: /si-akademik/api/mahasiswa/4
Content-Length: 289
Content-Type: application/json; charset=utf-8

{
    "status": "success",
    "message": "Mahasiswa berhasil ditambahkan",
    "data": {
        "id": 4,
        "nim": "2024004",
        "nama": "Dewi Lestari",
        "email": "dewi@mail.com",
        "jurusan": "Sistem Informasi",
        "created_at": "2026-10-20 21:44:30"
    }
}
```

3. Menambahkan data mahasiswa menggunakan method POST

```
PS C:\laragon\www\BKPM\Acara16> '{"nim":"2024001","nama":"Budi Santoso Update","email":"budi.baru@mail.com","jurusan":"Teknik Informatika"}' | Set-Content body.json -Encoding ascii -NoNewline
>> curl.exe -i -X PUT "$base/1" -H "Content-Type: application/json" --data-binary "@body.json"
HTTP/1.1 200 OK
Date: Tue, 20 Oct 2026 14:45:00 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Content-Length: 298
Content-Type: application/json; charset=utf-8

{
    "status": "success",
    "message": "Mahasiswa berhasil diubah",
    "data": {
        "id": 1,
        "nim": "2024001",
        "nama": "Budi Santoso Update",
        "email": "budi.baru@mail.com",
        "jurusan": "Teknik Informatika",
        "created_at": "2026-10-20 20:23:25"
    }
}
```

4. Mengubah data mahasiswa manggunakan Method PUT

```
PS C:\laragon\www\BKPM\Acara17> curl.exe -i -X DELETE "${base}?id=1"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 01:54:25 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 50
Content-Type: application/json

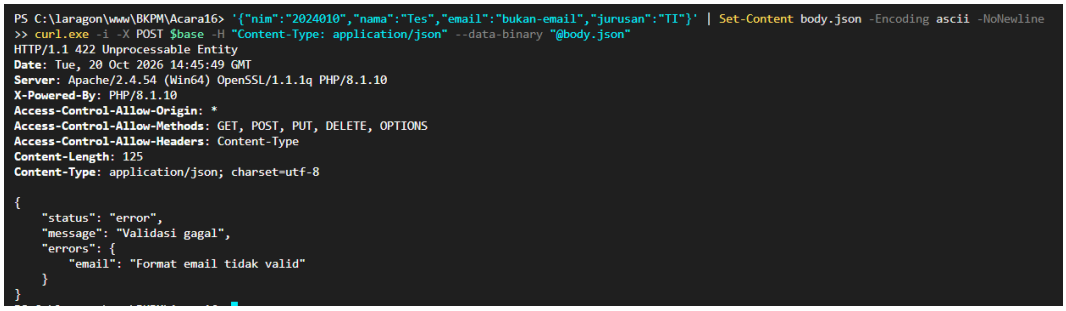
{"success":true,"message":"Data berhasil dihapus"}
PS C:\laragon\www\BKPM\Acara17>
```

5. Menghapus data mahasiswa dengan method DELETE

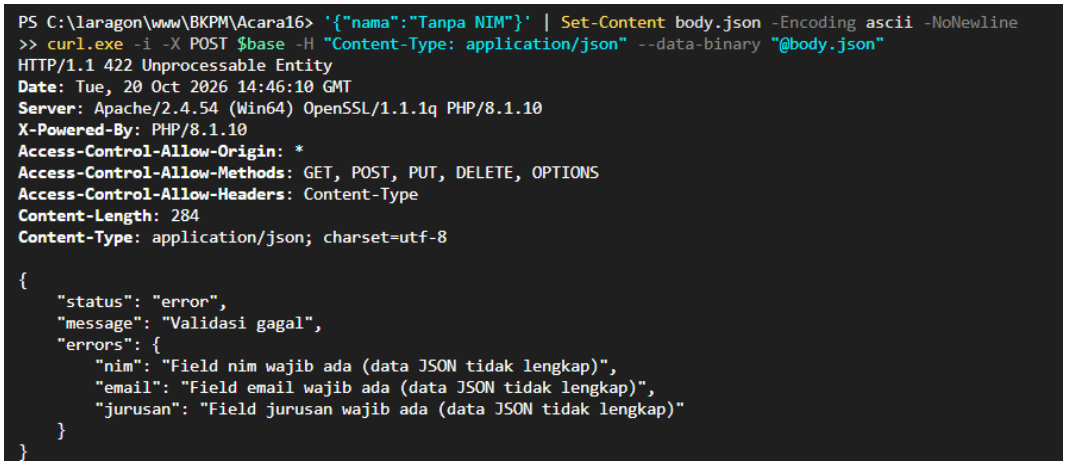
```
PS C:\laragon\www\BKPM\Acara16> '{"nim":"","nama":"Tes","email":"tes@mail.com","jurusan":"TI"}' | Set-Content body.json -Encoding ascii -NoNewline
>> curl.exe -i -X POST $base -H "Content-Type: application/json" --data-binary "@body.json"
HTTP/1.1 422 Unprocessable Entity
Date: Tue, 20 Oct 2026 14:45:28 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Content-Length: 127
Content-Type: application/json; charset=utf-8

{
    "status": "error",
    "message": "Validasi gagal",
    "errors": {
        "nim": "Field nim tidak boleh kosong"
    }
}
```

6. Contoh response error Nim Kosong



7. Contoh response error Format Email tidak valid



8. Contoh response error JSON tidak lengkap